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Description - This procedure describes the removal and replacement of the system support module (SSM), the analog power monitor (APM), the communications pin driver (CPD), the front panel display, the integrated power supply (IPS), and the rear panel assembly for the 1.0T and 1.5T RF/PEN II cabinet.

1- TOOLS REQUIRED

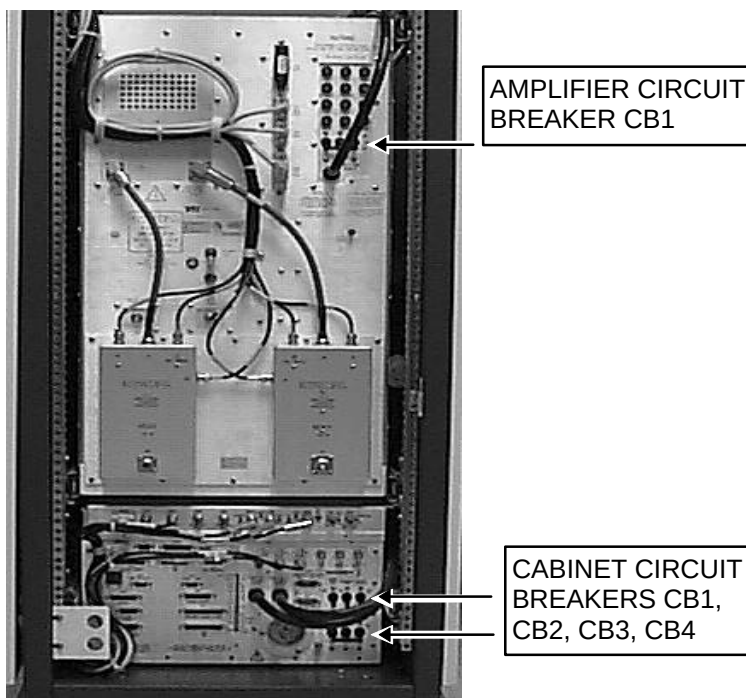
- #2 Phillips screwdriver
- #1 Phillips screwdriver
- 3/16" nut driver

WARNING!

POSSIBLE PERSONAL INJURY! REMOVE POWER FROM THE CABINET BEFORE REMOVING ANY COMPONENTS FROM THE SYSTEM SUPPORT MODULE. VERIFY THAT ALL BREAKERS IN THE REAR OF THE CABINET ARE OFF, AND DISCONNECT THE MAIN POWER FEED CORD BEFORE OPENING THE CHASSIS FOR SERVICE. SERIOUS INJURY OR DEATH BY ELECTROCUTION MAY OTHERWISE OCCUR. LOCK OUT AND TAG OUT THE CABINET.

2- REMOVAL OF ENTIRE SSM

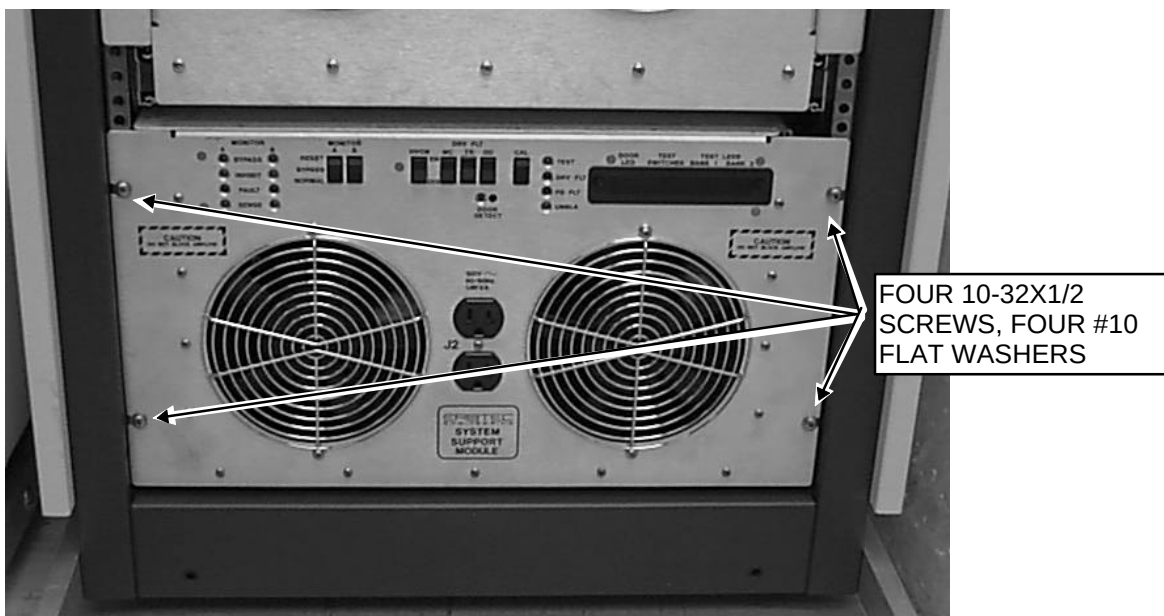
1. Turn off the cabinet power circuit breakers on the rear of the cabinet and RF amplifier. Lock out and tag out the cabinet (see Illustration 2-1). (Refer to *Procedure For Safety: Section 6.*)



LOCK OUT AND TAG OUT
ILLUSTRATION 2-1

2- REMOVAL OF ENTIRE SSM (continued)

2. Verify that power to the RF/Pen II Cabinet is off by checking the fans and LEDs on both the RF amplifier and the System Support Module (SSM).
3. Remove the front cover from the cabinet.
4. Remove the four 10-32x1/2 screws and four #10 flat washers that secure the SSM chassis to the cabinet (see Illustration 2-2).

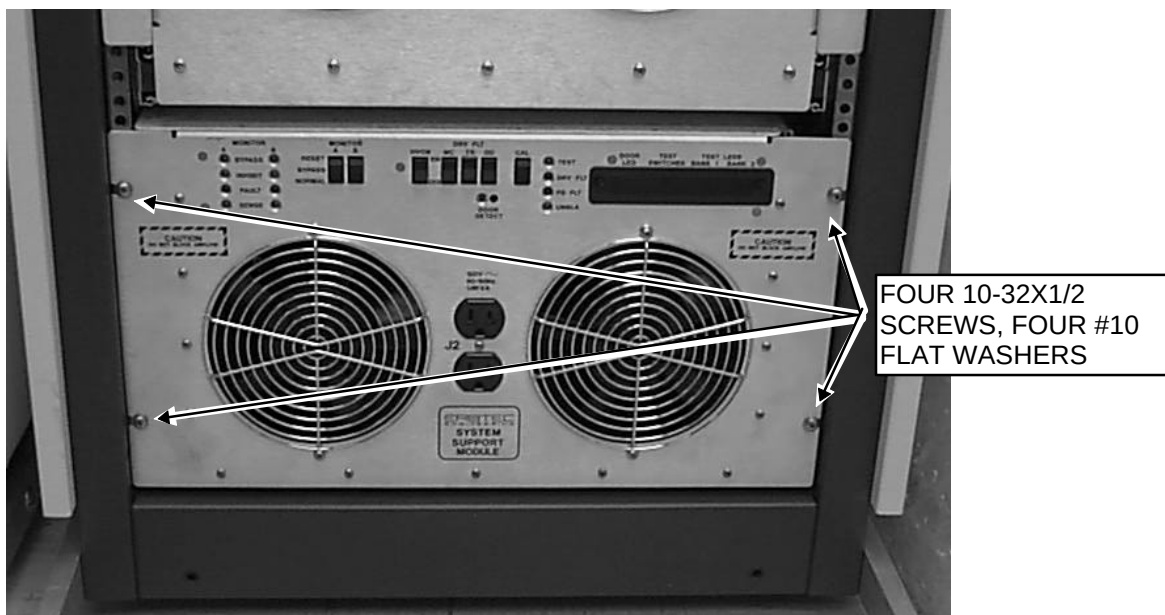


REMOVAL OF SYSTEM SUPPORT MODULE
ILLUSTRATION 2-2

5. Remove all external cabling to the SSM.
6. Slide out the SSM from the cabinet. Use care to ensure that external cables have sufficient slack and do not bind or pinch.
7. Support the module from beneath to alleviate pressure on the slides.
8. Remove the six 8-32x3/8 screws (three per slide) that secure the rack slides to the module. This enables you to completely remove the unit.

3- INSTALLATION OF NEW SSM

1. Support the SSM from beneath to alleviate pressure on the slides.
2. Install the six 8-32x3/8 screws (three per slide) that secure the rack slides to the module.
3. Carefully slide the SSM back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.)
4. Install the four 10-32x1/2 screws and four #10 flat washers that secure the SSM to the cabinet (see Illustration 3-1).

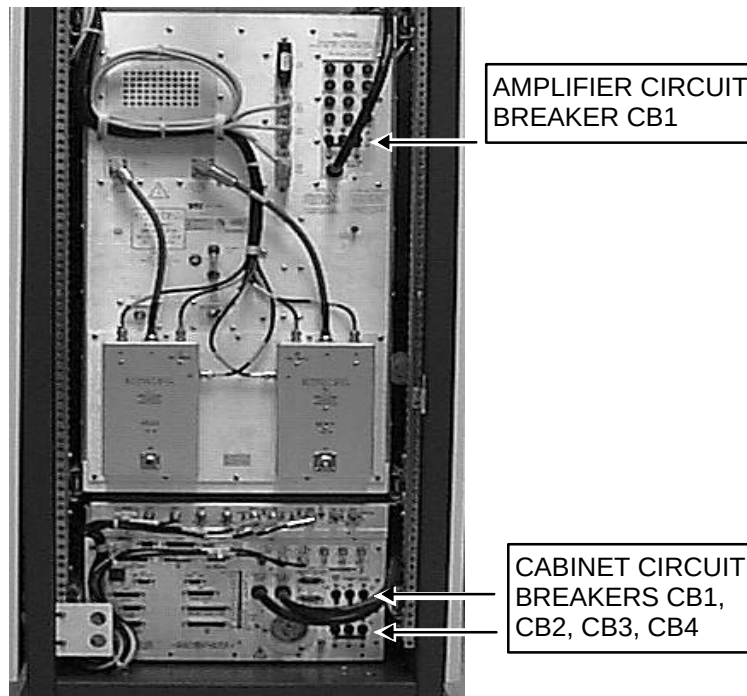


INSTALLATION OF SYSTEM SUPPORT MODULE
ILLUSTRATION 3-1

5. Replace all external connections.
6. Install front cover.

3- INSTALLATION OF NEW SSM (continued)

7. Remove lock out and tag out, and restore power to the cabinet (see Illustration 3-2).



LOCK OUT AND TAG OUT
ILLUSTRATION 3-2

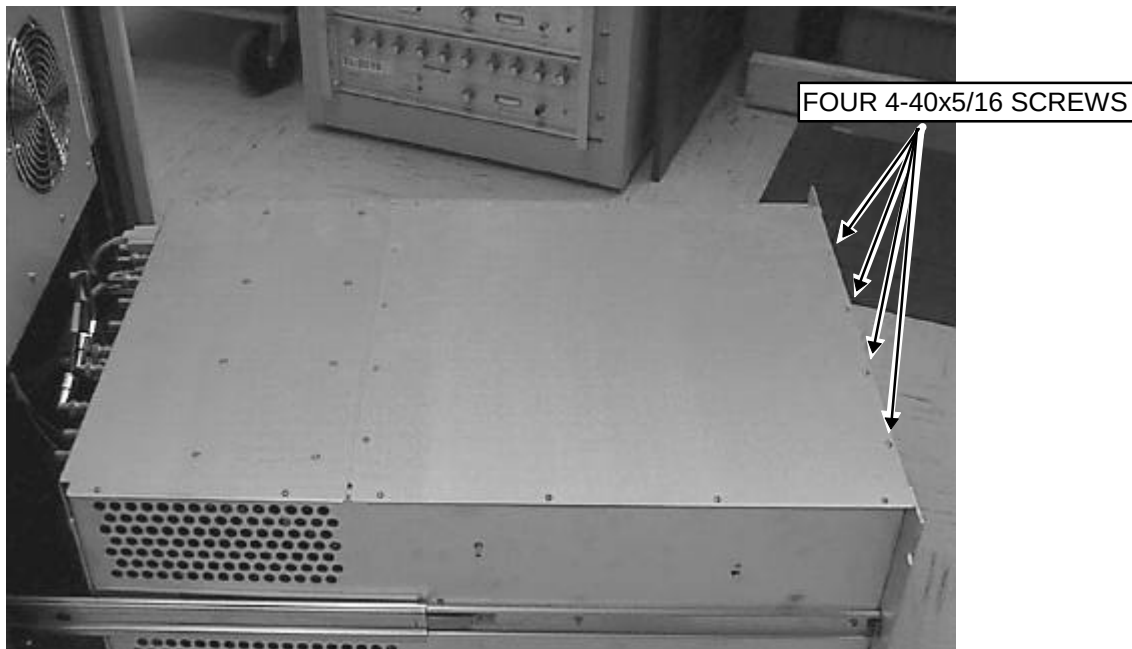
4- SLIDING OUT THE SSM

You will need to slide out the SSM in order to remove and replace any of its components.

1. Turn off the cabinet circuit breakers on the rear of the cabinet and RF amplifier. Lock out and tag out the cabinet (see Illustration 3-2). (Refer to *Procedure For Safety: Section 6.*)
2. Verify that power to the RF/Pen II Cabinet is off by checking the fans and LEDs on both the RF amplifier and the System Support Module (SSM).
3. Remove the front cover from the cabinet.
4. Remove the four 10-32x1/2 panhead screws and four #10 flat washers that secure the SSM chassis to the cabinet (see Illustration 3-1).
5. Slide out the SSM from the cabinet. Use care to ensure that external cables have sufficient slack and do not bind or pinch.

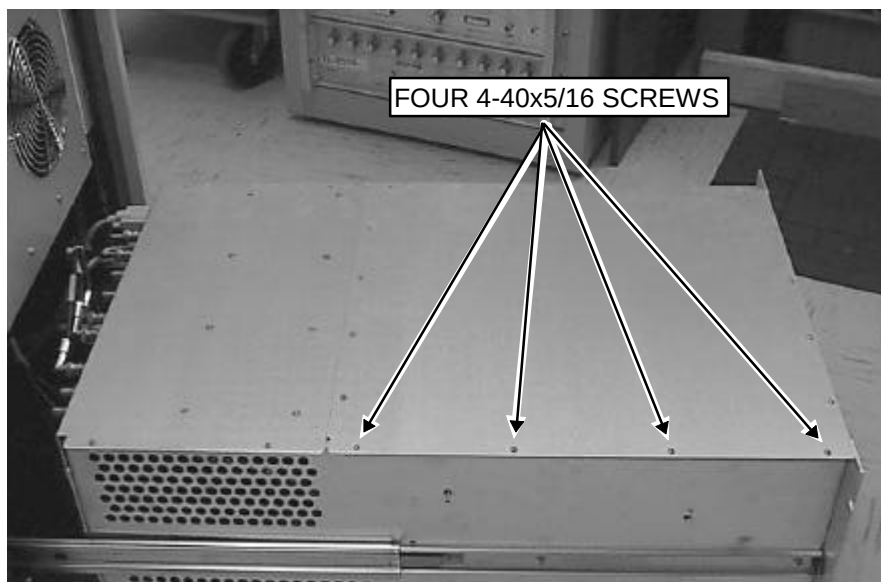
4- SLIDING OUT THE SSM (continued)

6. Remove the four 4-40x5/16 panhead screws that secure the front lid to the front panel (see Illustration 4-1).



REMOVAL OF SYSTEM SUPPORT MODULE LID FROM FRONT PANEL
ILLUSTRATION 4-1

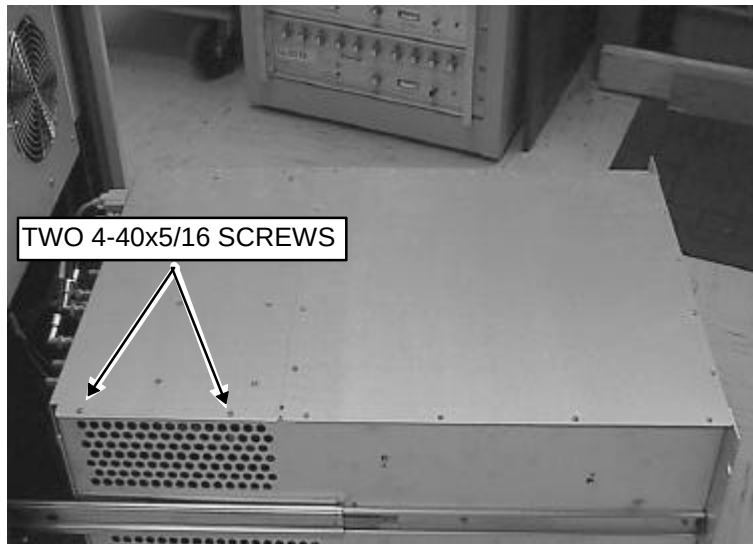
7. Remove the four 4-40x5/16 panhead screws that secure the front lid to the chassis (see Illustration 4-2).



REMOVAL OF SYSTEM SUPPORT MODULE LID FROM CHASSIS
ILLUSTRATION 4-2

4- SLIDING OUT THE SSM (continued)

8. Remove the two 4-40x5/16 panhead screws that secure the APM lid to the chassis (see Illustration 4-3).



REMOVAL OF ANALOG PROCESSOR MODULE LID FROM CHASSIS
ILLUSTRATION 4-3

9. Remove the three 4-40x5/16 panhead screws that secure the APM lid to the rear panel (see Illustration 4-4). A few cables may need to be disconnected first.



REMOVAL OF ANALOG POWER MONITOR LID FROM REAR PANEL
ILLUSTRATION 4-4

4- SLIDING OUT THE SSM (continued)

10. Grasp the edge of the module lid where the APM lid meets the front lid, and lift the lid to a vertical position. Use care to ensure that the APM cables do not catch on the bottom of the RF amplifier and that there is sufficient slack.

Go to the appropriate section to continue the remove and replace procedure:

See section 5- Removal of Failed APM Module

See section 7- Removal of Failed CPD Module

See section 9- Removal of Failed Front Panel Display Board

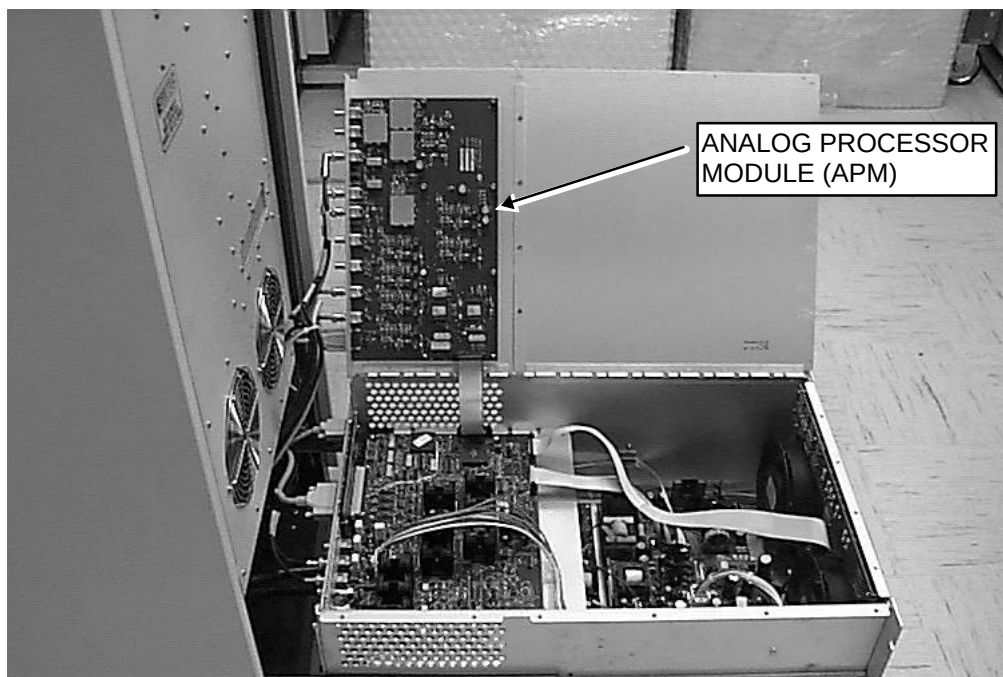
See section 11- Removal of Failed IPS Module

See section 13- Removal of Failed Rear Panel Assembly

5- REMOVAL OF FAILED APM MODULE

Verify that the power is off, and that the cabinet is locked out and tagged out.

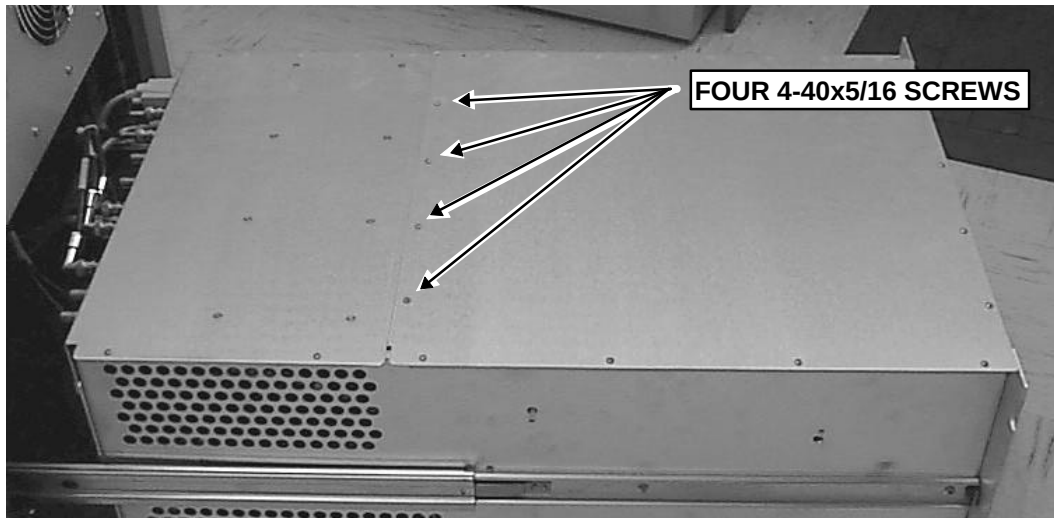
1. The APM module is the lid assembly near the rear of the SSM (see Illustration 5-1).



ANALOG PROCESSOR MODULE LOCATION
ILLUSTRATION 5-1

5- REMOVAL OF FAILED APM MODULE(continued)

2. Remove the four 4-40 x 5/16 panhead screws that secure the front lid to the APM lid (see Illustration 5-2).

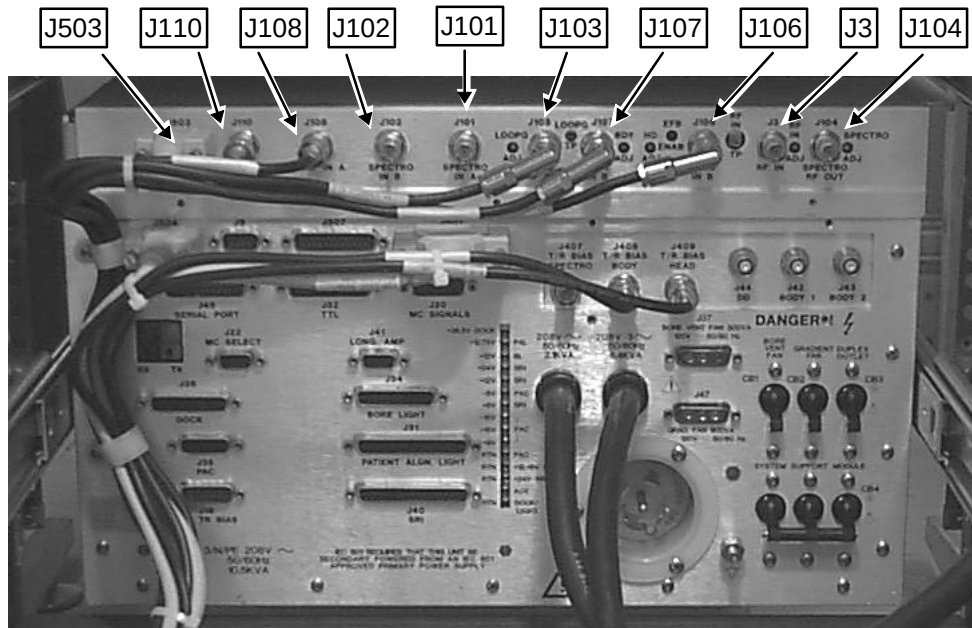


REMOVAL OF ANALOG PROCESSOR MODULE LID FROM FRONT LID
ILLUSTRATION 5-2

3. Disconnect the cables attached to the rear of the APM module and the ribbon cable attached to the APM board.
4. Remove the failed APM module by sliding lid forward in the upright position to disengage hinge.

6- INSTALLATION OF NEW APM MODULE

1. In the upright position, place the APM module on the hinges and slide to the back to attach.
2. Attach the ribbon cable from the CPD board to the APM board. Close APM lid.
3. Attach the cables to the APM board (see Illustration 6-1). Be sure to match the label designators on the cables to the APM connectors.



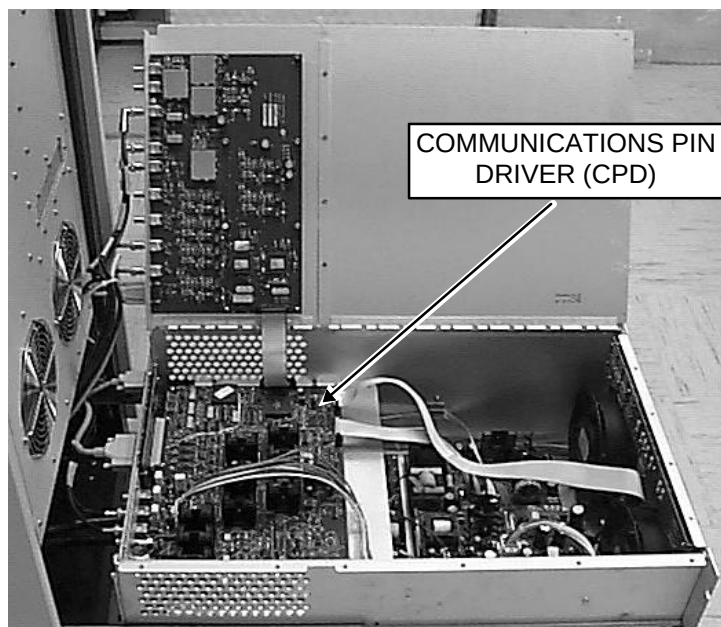
ANALOG PROCESSOR MODULE CABLE CONNECTIONS
ILLUSTRATION 6-1

4. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel.
5. Close the front lid. Install eight 4-40 x 5/16 panhead screws: four to the APM lid, and four to the front panel.
6. Attach six 4-40x5/16 panhead screws that secure both lids to chassis. Tighten all screws in steps 4, 5, and 6.
7. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.) Secure with four 10-32x1/2 panhead screws and four #10 flat washers.
8. Replace front cover.
9. Remove lock out and tag out, and restore power to the cabinet.

7- REMOVAL OF FAILED CPD MODULE

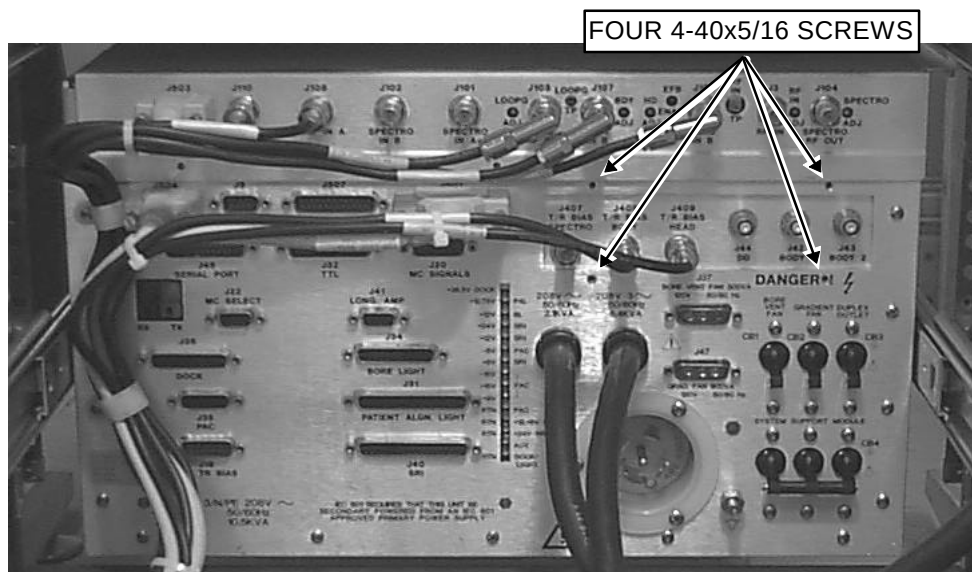
Verify that the power is off, and that the cabinet is locked out and tagged out.

1. The CPD module is located at the rear of the SSM (see Illustration 7-1).



COMMUNICATIONS PIN DRIVER LOCATION
ILLUSTRATION 7-1

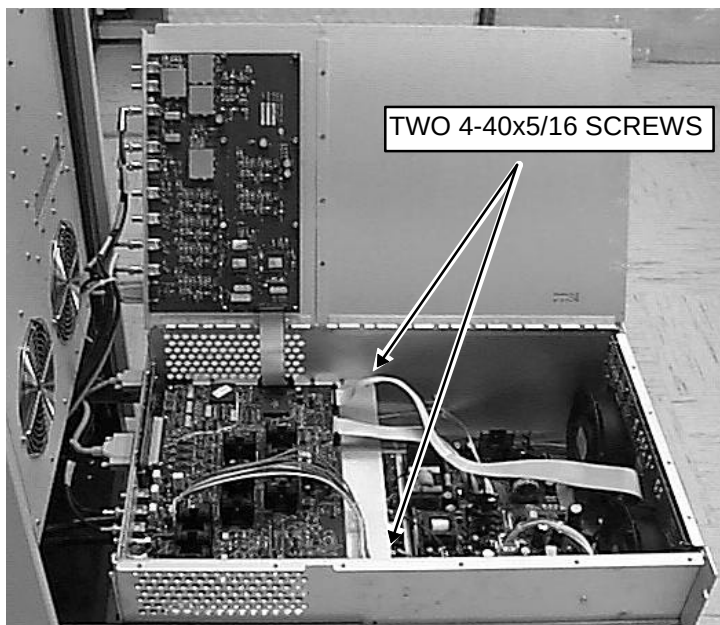
2. Disconnect all connectors attached to the CPD board and to the rear of the CPD module.
3. Remove four 4-40x5/16 panhead screws that secure the CPD module to the rear panel (see Illustration 7-2).



REMOVAL OF COMMUNICATIONS PIN DRIVER FROM REAR PANEL
ILLUSTRATION 7-2

7- REMOVAL OF FAILED CPD MODULE (continued)

4. Remove the two 4-40x5/16 panhead screws that secure the module to the side brackets (see Illustration 7-3).

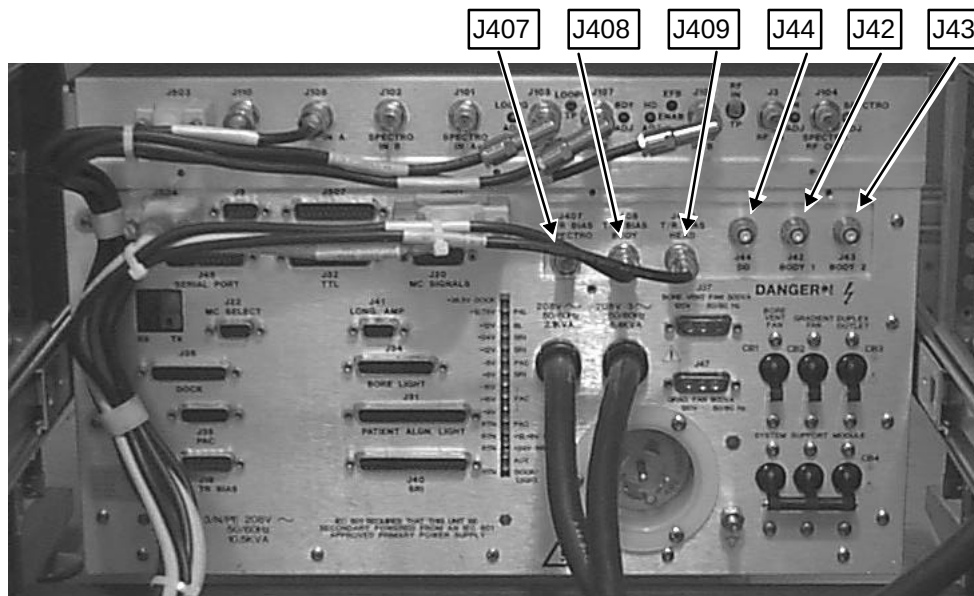


REMOVAL OF COMMUNICATIONS PIN DRIVER FROM SIDE BRACKETS
ILLUSTRATION 7-3

5. Slide CPD module toward the front of the SSM, carefully detaching DIN connector; lift and remove failed CPD module.

8- INSTALLATION OF NEW CPD MODULE

1. Place the CPD module on the brackets, aligning the slots, and slide toward the rear panel, carefully attaching the DIN connector to the rear panel.
2. Install four 4-40x5/16 screws through the outside of the rear panel into the CPD module.
3. Install two 4-40x5/16 panhead screws that secure the module to the side brackets. Tighten all screws in steps 2 and 3.
4. Attach all connections to the CPD board and to the rear of the CPD module (see Illustration 8-1).



COMMUNICATIONS PIN DRIVER CABLE CONNECTIONS

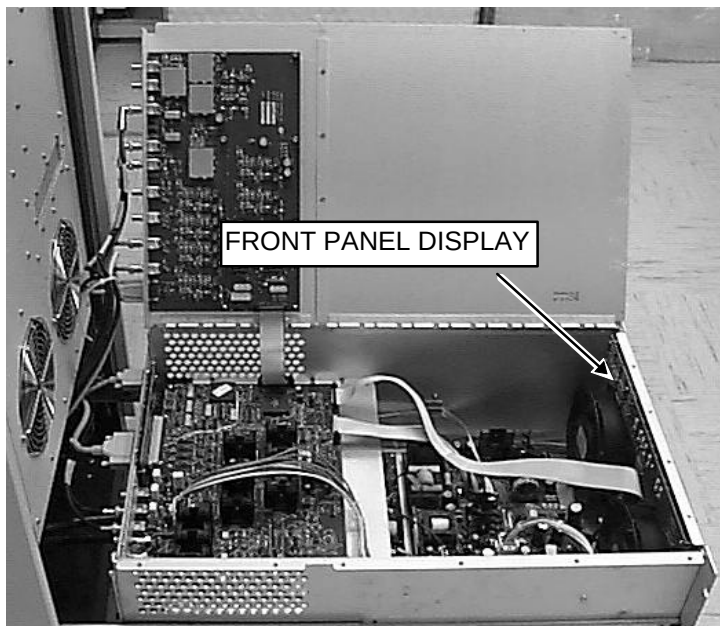
ILLUSTRATION 8-1

5. Close the module lids.
6. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel.
7. Install four 4-40x5/16 panhead screws that secure the front lid to the front panel.
8. Install six 4-40x5/16 panhead screws that secure both lids to the chassis. Tighten all screws in steps 6, 7, and 8.
9. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.) Secure with four 10-32x1/2 panhead screws and four #10 flat washers.
10. Replace the front cover.
11. Remove lock out and tag out, and restore power to the cabinet.

9- REMOVAL OF FAILED FRONT PANEL DISPLAY BOARD

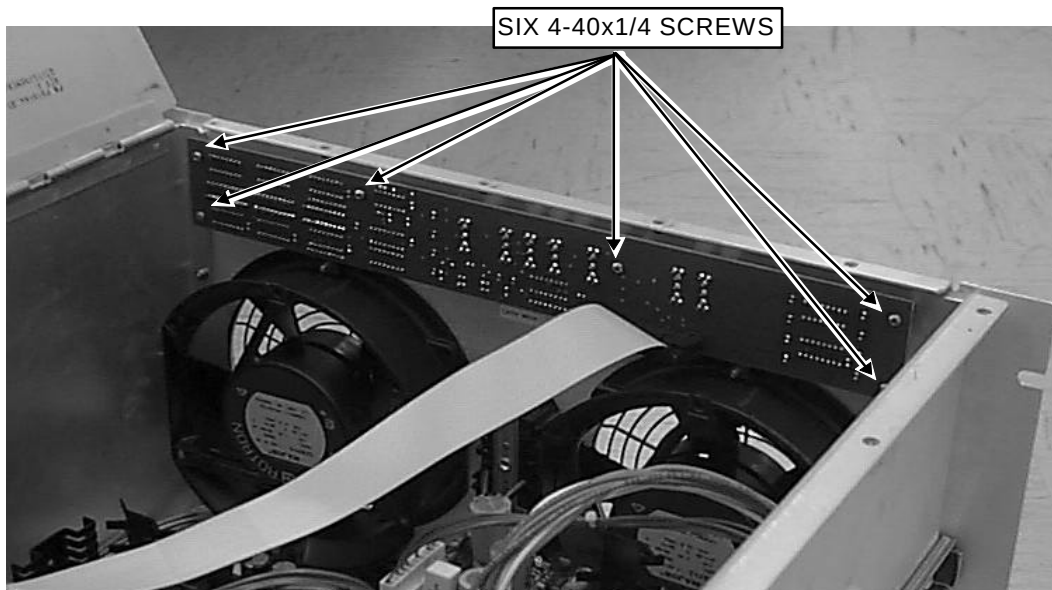
Verify that the power is off, and that the cabinet is locked out and tagged out.

1. The front panel display board is located on the front panel (see Illustration 9-1).



FRONT PANEL DISPLAY LOCATION
ILLUSTRATION 9-1

2. Disconnect the connector attached to the front panel display board.
3. Remove six 4-40x1/4 panhead screws that attach the front panel display board to front panel (see Illustration 9-2).



REMOVAL OF FRONT PANEL DISPLAY
ILLUSTRATION 9-2

4. Remove the failed front panel display board.

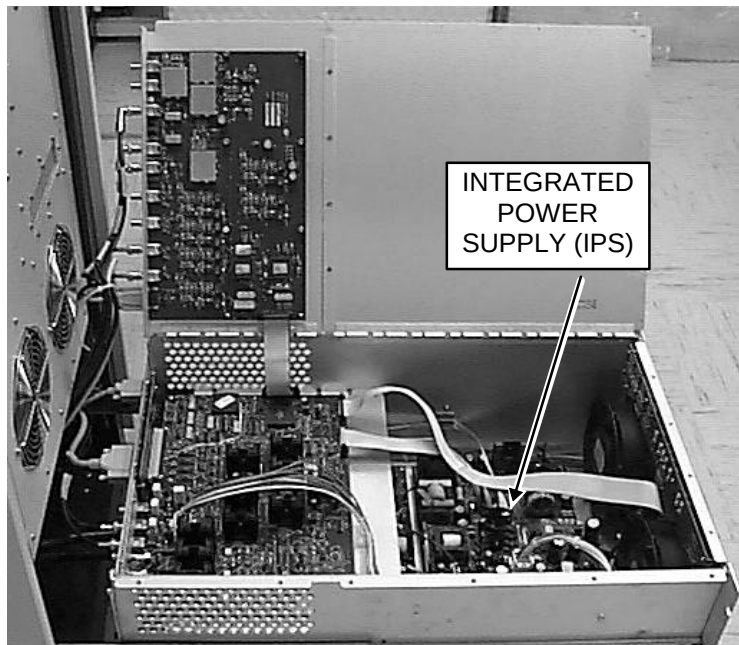
10- INSTALLATION OF NEW FRONT PANEL DISPLAY BOARD

1. Install the front panel display board on front panel.
2. Install and tighten six 4-40x1/4 panhead screws that secure the board to the front panel.
3. Attach the connector from the CPD board to the front panel display board. Close lids.
4. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel.
5. Install four 4-40x5/16 panhead screws that secure the front lid to the front panel.
6. Install six 4-40x5/16 panhead screws that secure both lids to chassis. Tighten all screws in steps 4, 5, and 6.
7. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.) Secure with four 10-32x1/2 panhead screws and four #10 flat washers.
8. Replace front cover.
9. Remove lock out and tag out, and restore power to the cabinet.

11- REMOVAL OF FAILED IPS MODULE

Verify that the power is off, and that the cabinet is locked out and tagged out.

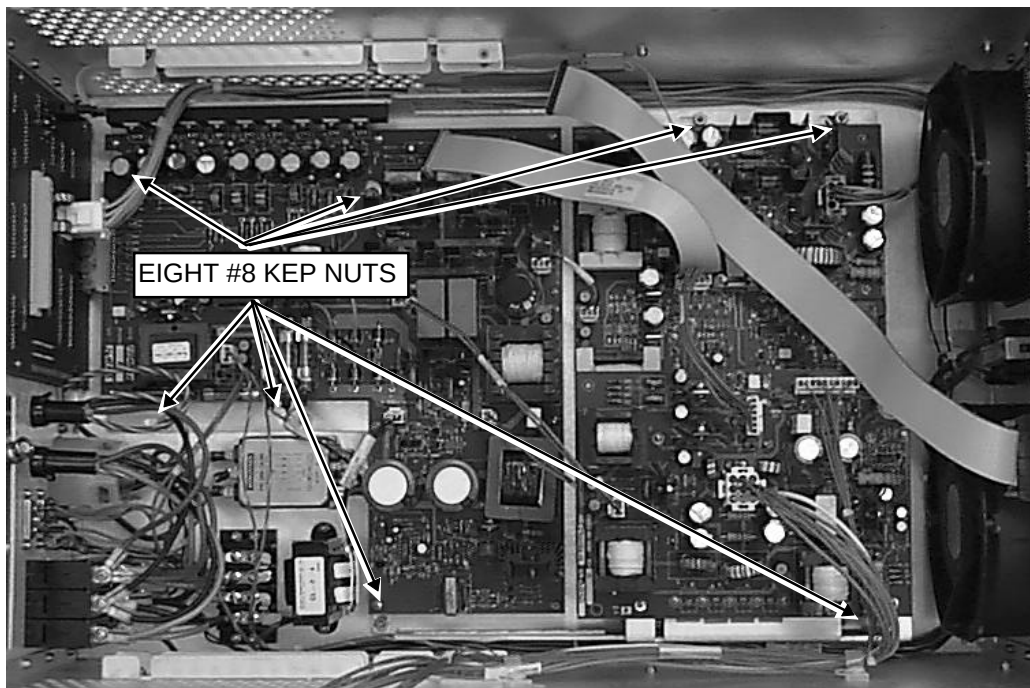
1. The IPS module is located on the bottom of the SSM (see Illustration 11-1).



INTEGRATED POWER SUPPLY LOCATION
ILLUSTRATION 11-1

11- REMOVAL OF FAILED IPS MODULE(continued)

2. The CPD module will have to be removed before the IPS can be removed. See section 7- Removal of Failed CPD Module for instructions on removing the CPD.
3. Disconnect all cables attached to the IPS boards, except those that are internal to the IPS assembly.
4. Using a nut driver, remove the eight #8 KEP nuts that secure the IPS bulkhead to the bottom of the chassis (see Illustration 11-2).



REMOVAL OF INTEGRATED POWER SUPPLY
ILLUSTRATION 11-2

5. Gently pull the tray up and toward the front in order to disengage the DIN connector on the rear panel while clearing the alignment pins.
6. Lift the entire IPS module out of the chassis.

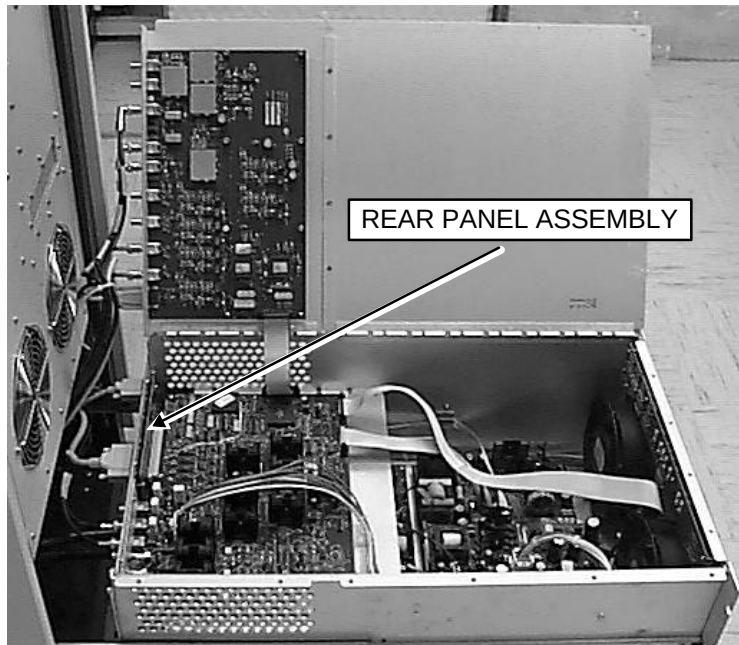
12- INSTALLATION OF NEW IPS MODULE

1. Make sure the CPD module has been removed before trying to replace the IPS module.
2. Carefully lower the IPS module into the chassis, aligning the bulkhead slots with the chassis standoffs, and engaging the DIN connector on the rear panel.
3. Secure the module to the chassis with eight #8 KEP nuts.
4. Connect all cabling to the IPS module.
5. Replace the CPD module as in section 8.
6. Close the module lids.
7. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel.
8. Install four 4-40x5/16 panhead screws that secure the front lid to the front panel.
9. Install six 4-40x5/16 panhead screws that secure both lids to the chassis. Tighten all screws in steps 7, 8, and 9.
10. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.) Secure with four 10-32x1/2 panhead screws and four #10 flat washers.
11. Replace the front cover.
12. Remove lock out and tag out, and restore power to the cabinet.

13- REMOVAL OF FAILED REAR PANEL ASSEMBLY

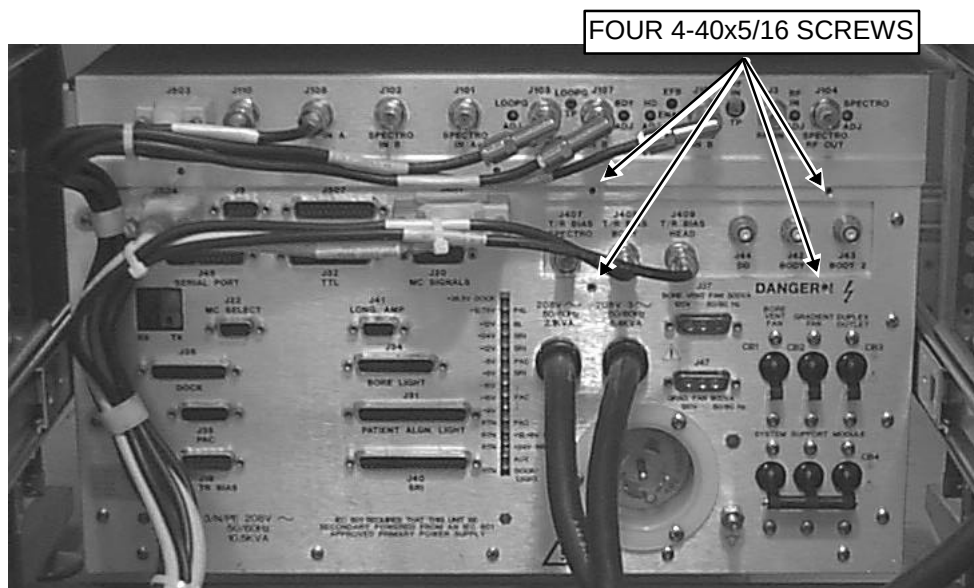
Verify that the power is off, and that the cabinet is locked out and tagged out.

1. The rear panel assembly is located on the rear of the SSM (see Illustration 13-1).



REAR PANEL ASSEMBLY LOCATION
ILLUSTRATION 13-1

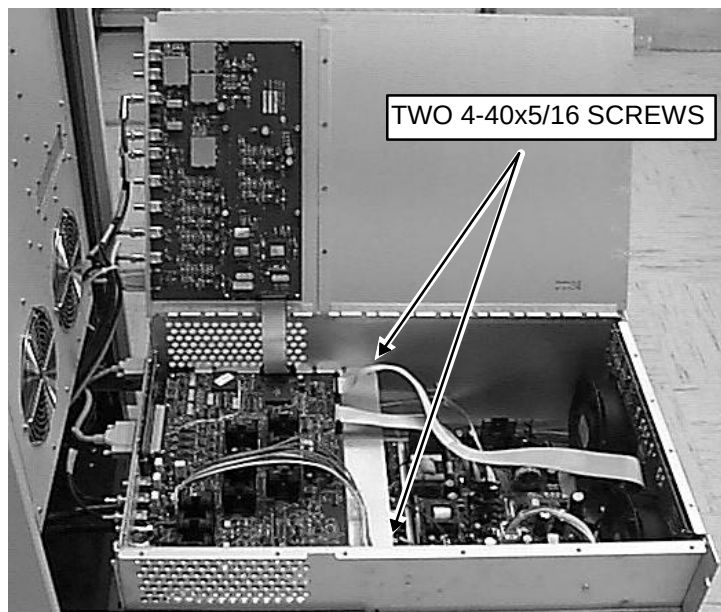
2. Disconnect all cables on the outside of the rear panel, except for those connected to the APM.
3. Remove four 4-40x5/16 panhead screws that secure the CPD module to the rear panel (see Illustration 13-2).



REMOVAL OF COMMUNICATIONS PIN DRIVER FROM REAR PANEL
ILLUSTRATION 13-2

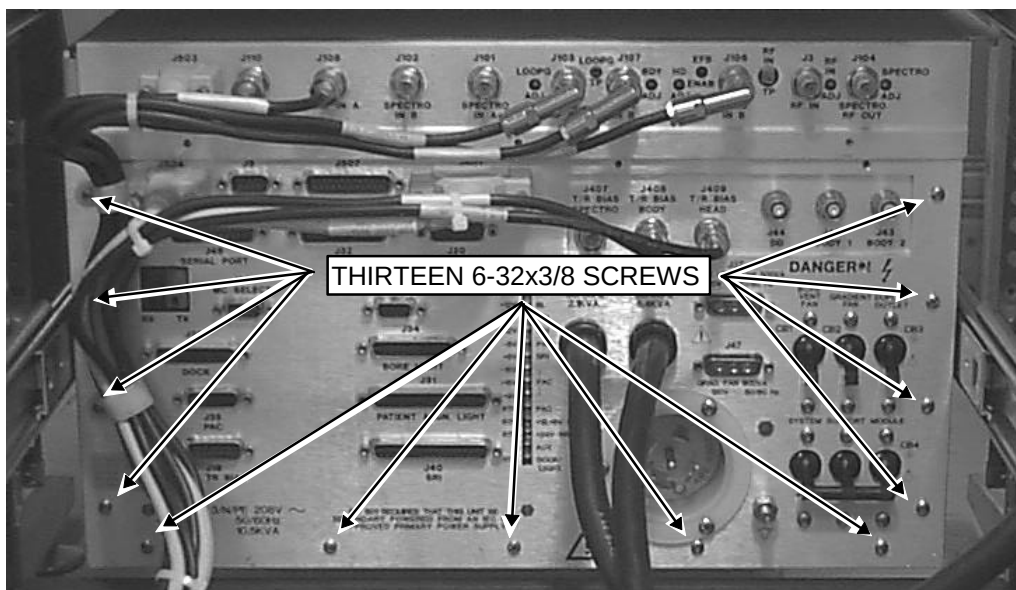
13- REMOVAL OF FAILED REAR PANEL ASSEMBLY (continued)

4. Remove the two 4-40x5/16 panhead screws that secure the CPD module to the side brackets (see Illustration 13-3).



REMOVAL OF COMMUNICATIONS PIN DRIVER FROM SIDE BRACKETS
ILLUSTRATION 13-3

5. Slide the CPD module toward the front of the SSM, carefully detaching the DIN connector.
6. Disconnect all internal cables to the rear interface board, including the DIN connector on the IPS module.
7. Remove thirteen 6-32x3/8 panhead screws that secure the rear panel to the chassis (see Illustration 13-4).



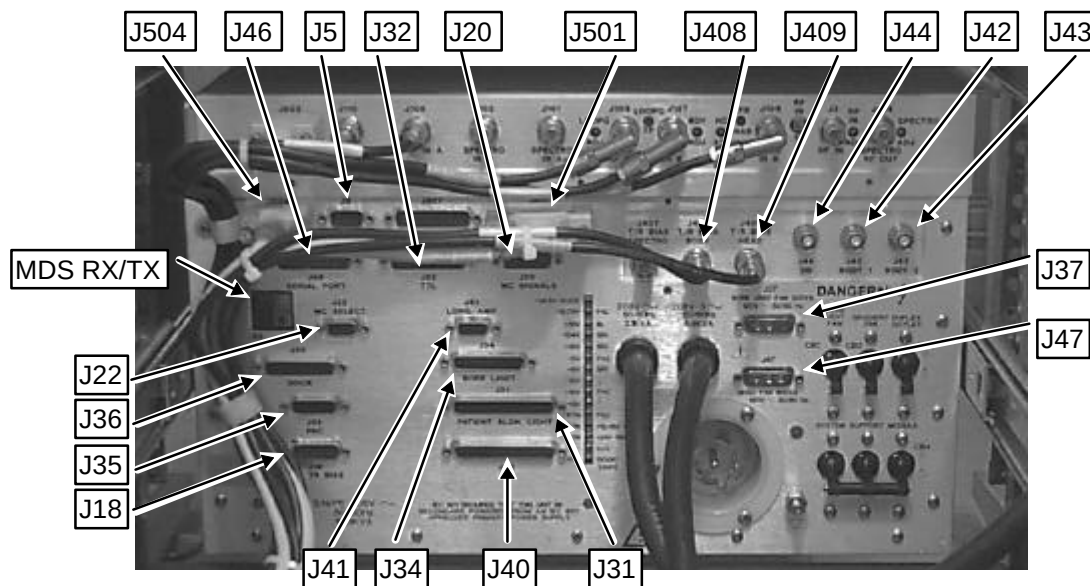
REMOVAL OF REAR PANEL ASSEMBLY
ILLUSTRATION 13-4

13- REMOVAL OF FAILED REAR PANEL ASSEMBLY (continued)

8. Remove one 4-40x5/16 panhead screw on the front panel that secures the duplex connector, and disconnect power to the fans. Release the harness from all cable clamps.
9. Remove the failed rear panel assembly and the associated cable harnesses.

14- INSTALLATION OF NEW REAR PANEL ASSEMBLY

1. Install thirteen 6-32x3/8 panhead screws that secure the rear panel to the chassis.
2. Install one 4-40x5/16 panhead screw on the front panel that secures the duplex connector, and connect power to the fans. Attach all cable clamps to the harness.
3. Connect all internal cables to the rear interface board, including the DIN connector on the IPS module.
4. Slide the CPD board toward the rear panel, and plug in the DIN connector. Attach four 4-40x5/16 panhead screws that secure the CPD board to the rear panel. Attach two 4-40x5/16 panhead screws that secure the CPD module to the side brackets.
5. Attach all external connections (see Illustration 14-1).



REAR PANEL ASSEMBLY CONNECTIONS
ILLUSTRATION 14-1

6. Close the module lids.
7. Install three 4-40x5/16 panhead screws that secure the APM lid to the rear panel.
8. Install four 4-40x5/16 panhead screws that secure the front lid to the front panel.
9. Install six 4-40x5/16 panhead screws that secure both lids to the chassis. Tighten all screws in steps 6, 7, and 8.

14- INSTALLATION OF NEW REAR PANEL ASSEMBLY (continued)

10. Slide the module back into the cabinet. (Press the slide locks to enable the module to slide completely into the cabinet.) Secure with four 10-32x1/2 panhead screws and four #10 flat washers.
11. Replace front cover.
12. Remove lock out and tag out, and restore power to the cabinet.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 14, 1998	Erbtec Engineering	Initial Toolbook conversion to Word. (KK)
1	May 21, 1999	SM Atladottir	Updated Procedure References for New GUI